

GRAiN *Oryza* - Genetic diversity, Resilience and Adaptation in Native *Oryza* of the Americas



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Introduction

The evolutionary histories of cultivated rice (*Oryza sativa* and *O. glaberrima*) and their wild relatives from Asia and Africa have been extensively described (Figure 1). On the contrary, the wild rice species endemic to the American continent are still understudied.

To fill this gap, the overarching goal of GRAiN *Oryza* is to understand diversity, evolutionary history, and useful genetic variation for climate resilience of four wild *Oryza* species from the Americas: *O. alta* Swallen (CCDD), *O. grandiglumis* (Doell) Prod (CCDD), *O. latifolia* Desv (CCDD) and *O. glumipatula* Steud (AA) (Vaughan DA, 1989).

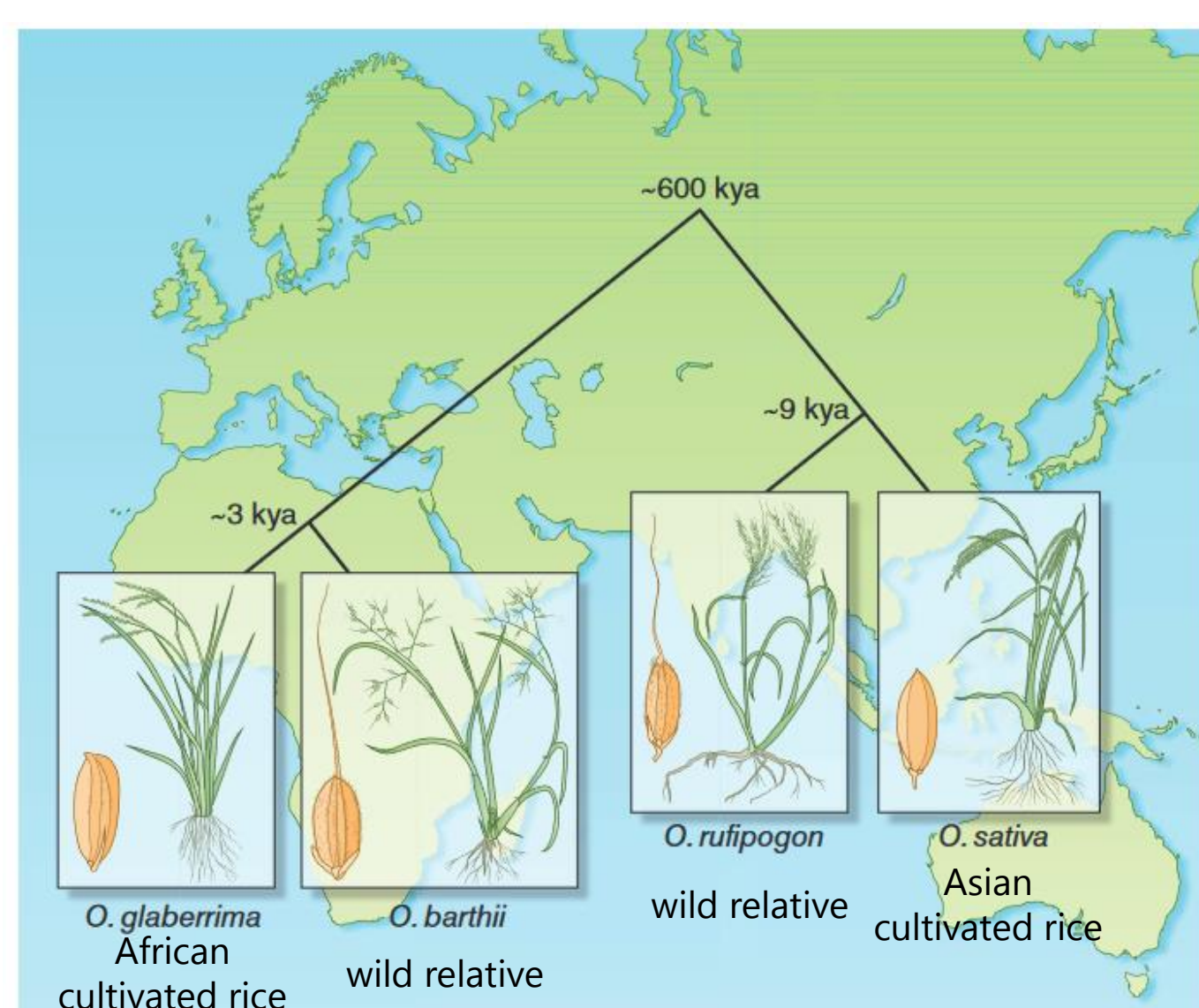


Figure 1. Parallel domestications of two rice species. *O. sativa* and *O. glaberrima* were domesticated in parallel from wild *O. rufipogon* in Asia and wild *O. barthii* in Africa, respectively. Modified from Purugganan MD. *Nat Genet.* (2014).

Goals

We generated a resequencing dataset of living and historical specimens of the four wild *Oryza* species collected across a wide geographic range (from Mexico to North Argentina) to address the following questions:

1. **Explore spatial and temporal genetic diversity.** How much diversity can be found in past and contemporary populations, and how much has it changed over time and space?
2. **Understand origin and evolutionary history of *O. glumipatula*.** Has this species been pre-domesticated and then returned to its wild state?
3. **Investigate genetic adaptation in *O. latifolia*.** Does this widely distributed species host ecological adaptations driving resilience across different environments?

Spatial and temporal range of the resequencing dataset

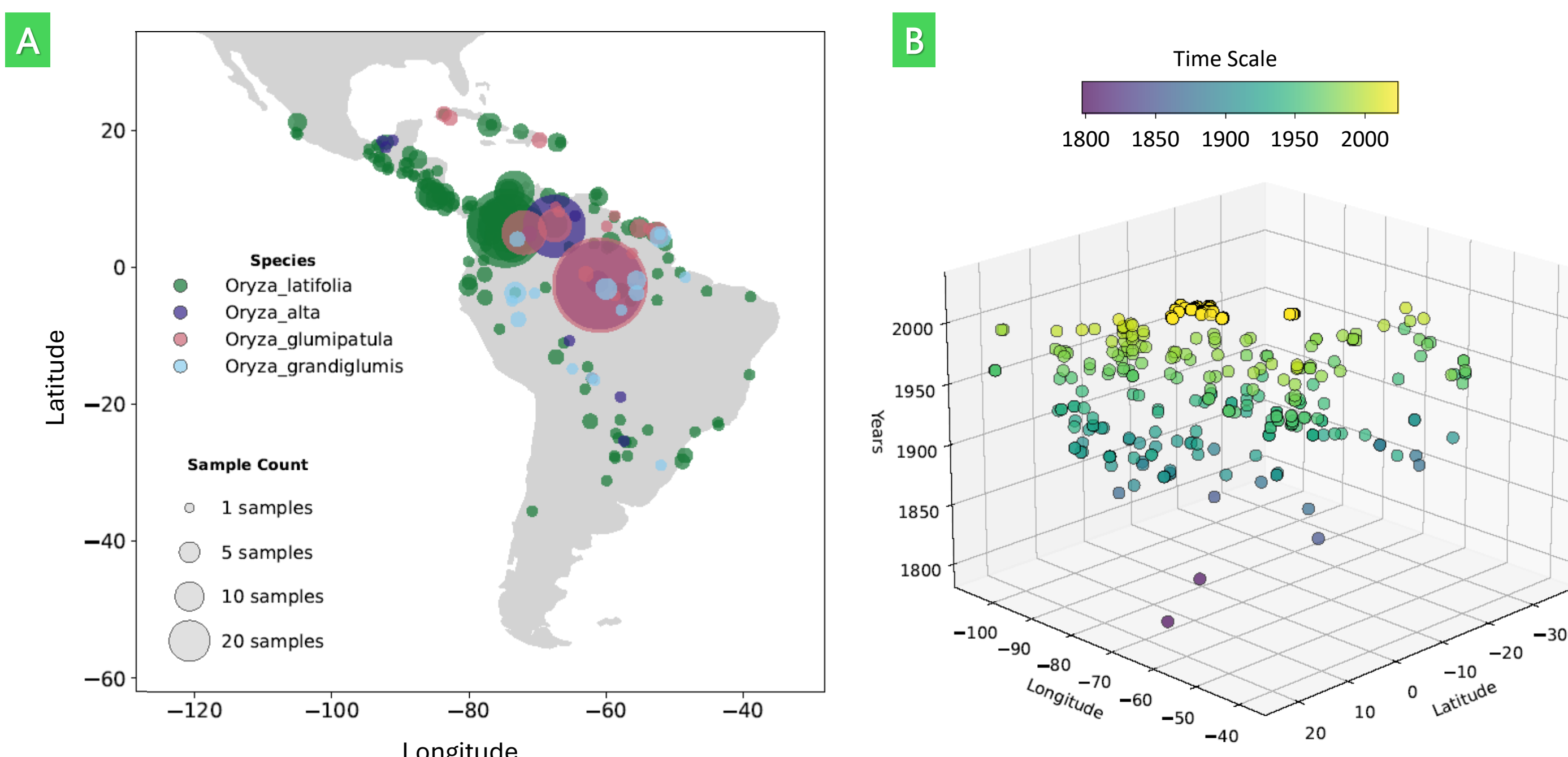


Figure 2. A) Map showing the geographic distribution and density of the historical and living specimens under study by species; and B) 3-D map showing both spatial and temporal distribution of the historical and living specimens under study by year of collection.

We isolated ancient DNA (aDNA) from 321 historical specimens and modern DNA from 459 living specimens (Figure 2), and prepared libraries for NGS sequencing. We included high-depth coverage sequencing of the type specimens for each species. Sequencing data from 9 samples was analysed to check authenticity of aDNA (see next panel).

Authentication of aDNA sequencing data

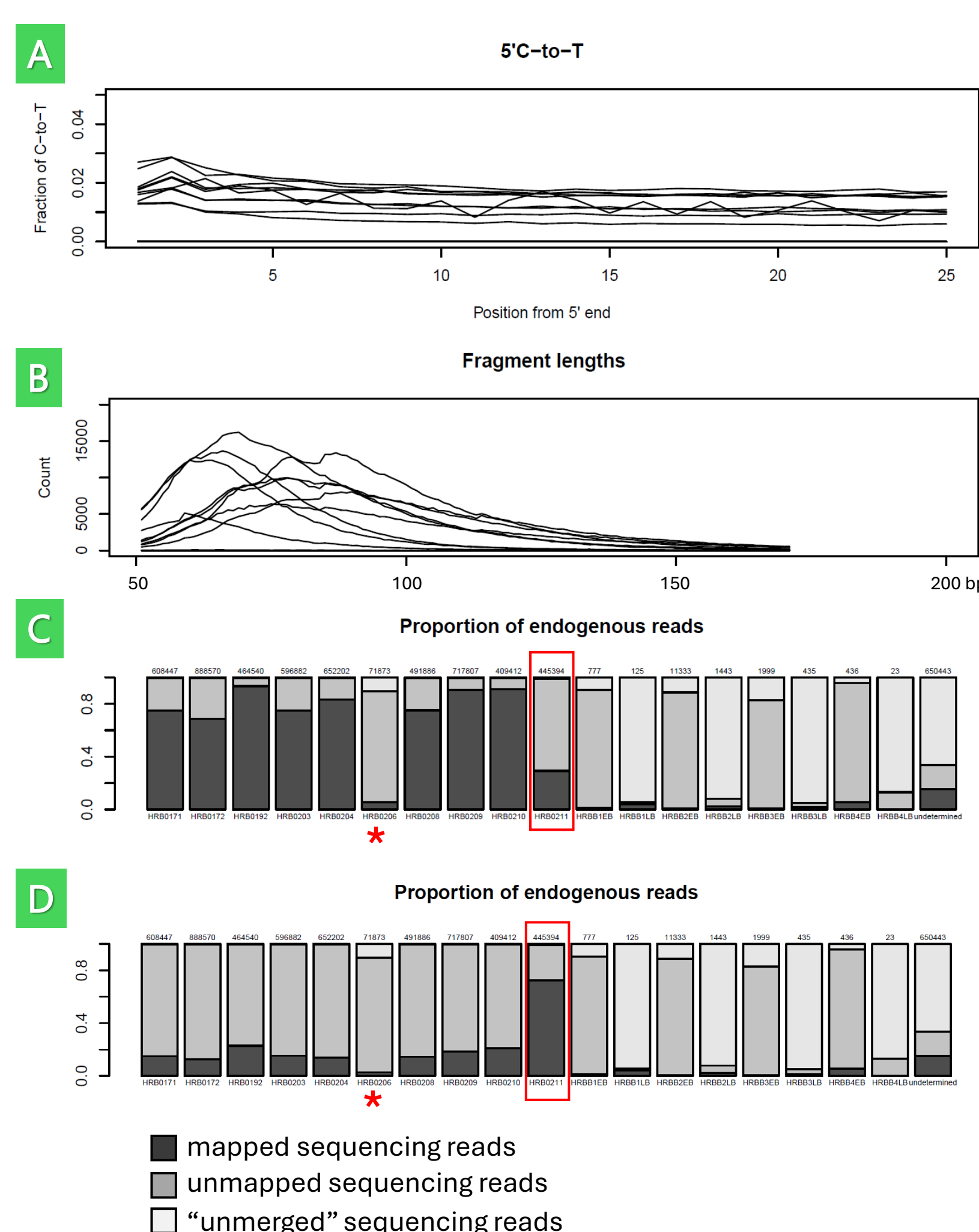


Figure 3. QC analysis of aDNA sequencing data.

A) Samples show 2-4% of C-to-T substitution at the 5' end of the sequencing reads. This method allows to authenticate aDNA samples.

B) The distribution of library fragment length shows a median value of around 70 – 90 bp, as expected from plant aDNA (Gutaker and Burbano, 2017).

C-D) Sequencing reads are mapped against *O. latifolia* (C) and *O. sativa* (D) reference genome sequences, respectively. Most of the samples (HRB0XXX) have around 80% of endogenous DNA (exogenous DNA is usually microbial DNA from microorganisms present *ante-mortem* or colonizing the tissue *post-mortem*).

Samples labelled as HRBXEB and HRBXLB (Figure 3C-D) are negative controls of the extraction and library preparation steps, respectively. Negative controls show <5 % of rice DNA, meaning the sequencing was relatively free of contamination.

Sample HRB206 (asterisk) shows very low percentage (<10%) of endogenous DNA.

Sample HRB211 (rectangle) most likely is cultivated *O. sativa* that was mislabeled by the botanist at the time of collection.

Puzzling questions on domesticated rice in the Americas

- Has rice ever been domesticated in the Americas?
- Is *O. glumipatula* a recent escape from an introduced rice?

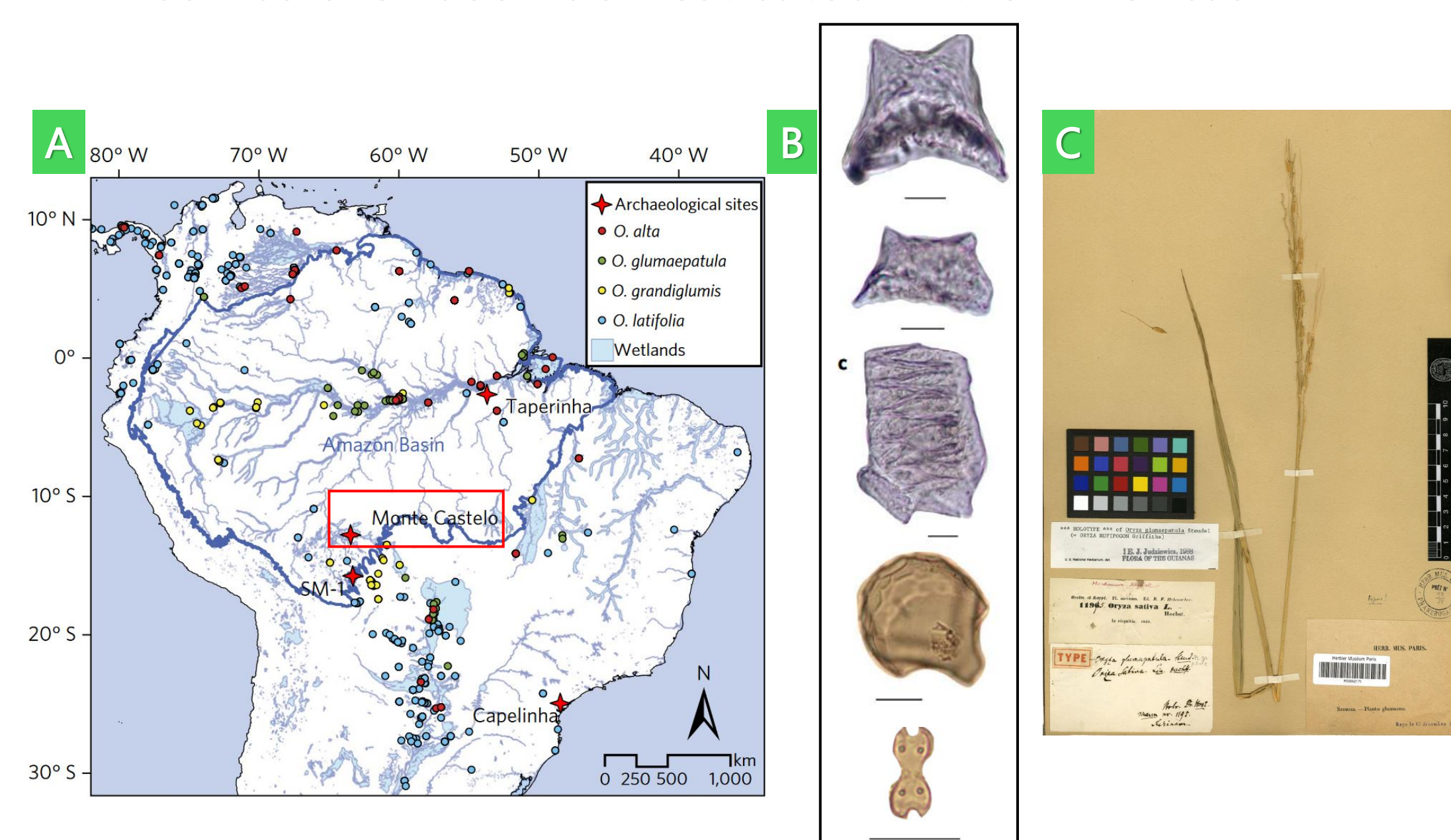
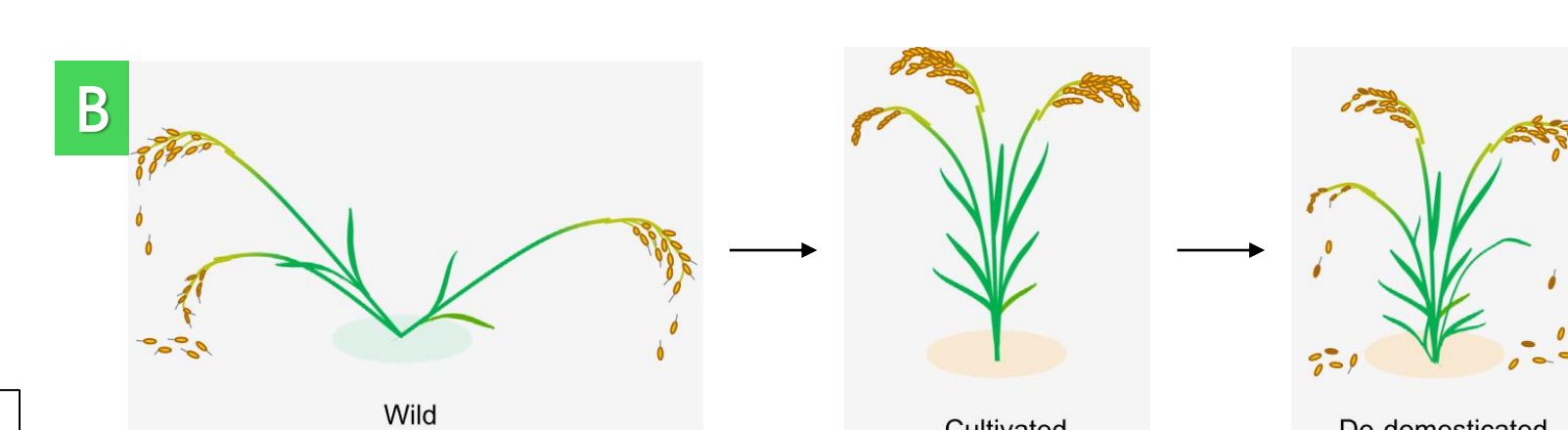


Figure 4. Evidence for rice domestication in the Americas. A) Monte Castelo archaeological site (Brazil). B) *Oryza* sp. husk phytolith morphotypes recovered at Monte Castelo. Increasingly larger phytoliths recovered from mid-Holocene levels of a shell mound dating to ca. 4000 cal yr BP show the progressive selection of larger wild rice seeds by its pre-Columbian residents. Modified from Hilbert L et al. *Nat Ecol Evol.* (2017). C) *O. glumipatula* type specimen collected by FW Hostmann in 1840 and described as cultivated in Surinam.

Figure 5. Rice introduction in the Americas from the Old World. A) Rice cultivation in the Americas by enslaved Africans (16th-19th century). Source: b-womeninamericanhistory19.blogspot.com



B) The de-domestication (or feralization) process involves the escape of cultivars from the farming environment through the acquisition of progressive independence from human intervention for dispersal and reproduction. Modified from Wu D, Lao S, Fan L. *Trends Plant Sci.* (2021).

Acknowledgments

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